Exploratory Data Analysis Memo

C22 Retail Fade Strategy

Karthik Manda | NTU veNTUre Programme

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0. Initial Hypotheses

Prior to formal testing, four behavioral hypotheses were formed from the challenge rules and kickoff briefing:

- **H1 — Cost dominance.** Given C22's stated ∼60% baseline fade win rate and the challenge's tight drawdown/target structure, trader net losses were expected to be driven substantially by transaction costs rather than pure directional error.
- **H2 — Loss-triggered tilt.** Consecutive losing trades were expected to predict deteriorating subsequent trade quality (revenge trading, size escalation, stop-loss abandonment), consistent with the "known behavioral biases" the brief describes.
- **H3 — Rule-boundary drift.** Traders were expected to drift toward the consistency-rule boundaries (2.5× lot cliff, 4% drawdown) under time and P&L pressure within the 24-hour window.
- **H4 — Rule design shapes behavior.** Per the briefing's framing that "the rules of a system change how people behave inside it," campaign-level parameter or platform changes were expected to be visible as discontinuities in behavioral statistics.

Sections 1–4 test these in turn: H1 is confirmed and quantified (Section 1); H2 is confirmed for the `loss_streak >= 2` family (Section 3) but not for all tilt proxies tested; H3 was tested via the `rule_b_breach` and ATLSR-proximity features but did not reach significance at current sample size; H4 is confirmed via the Campaign 53 discontinuity (Section 2).

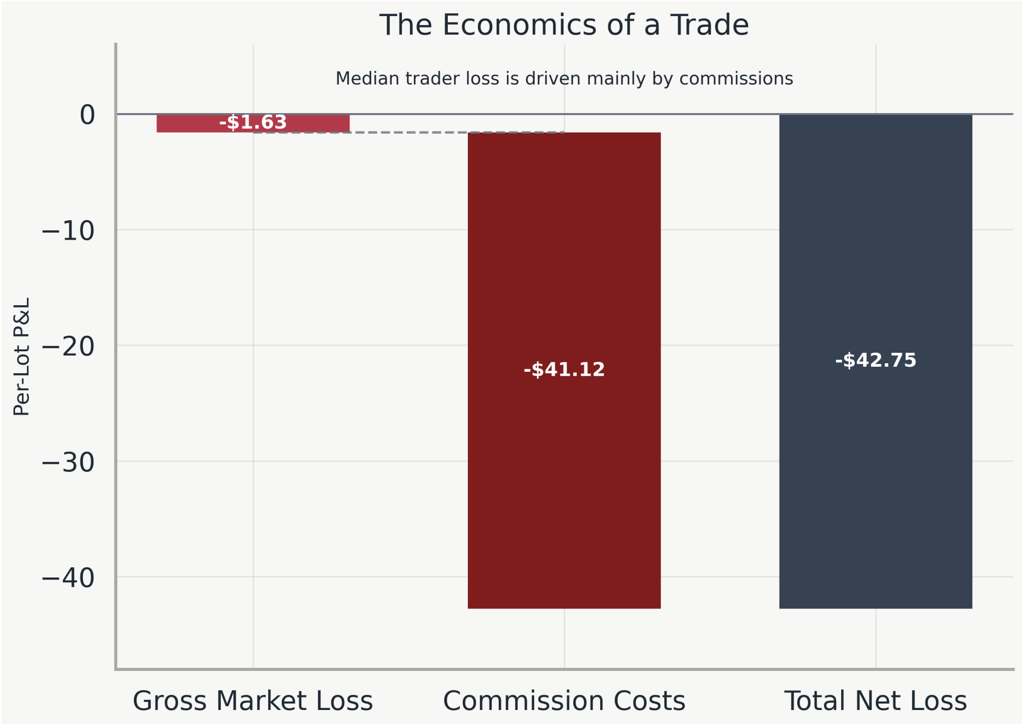
1. Structural Findings: The Economics of a Trade

The gross-basis fade hurdle is deterministic. The firm's profit from fading a retail trader's position follows an exact formula, confirmed to hold across every campaign in the corpus (min. $6.50/lot, on the single excluded test campaign):

$$\texttt{reverseProfit} = -\texttt{profit} - (7.00 \times \texttt{amount})$$

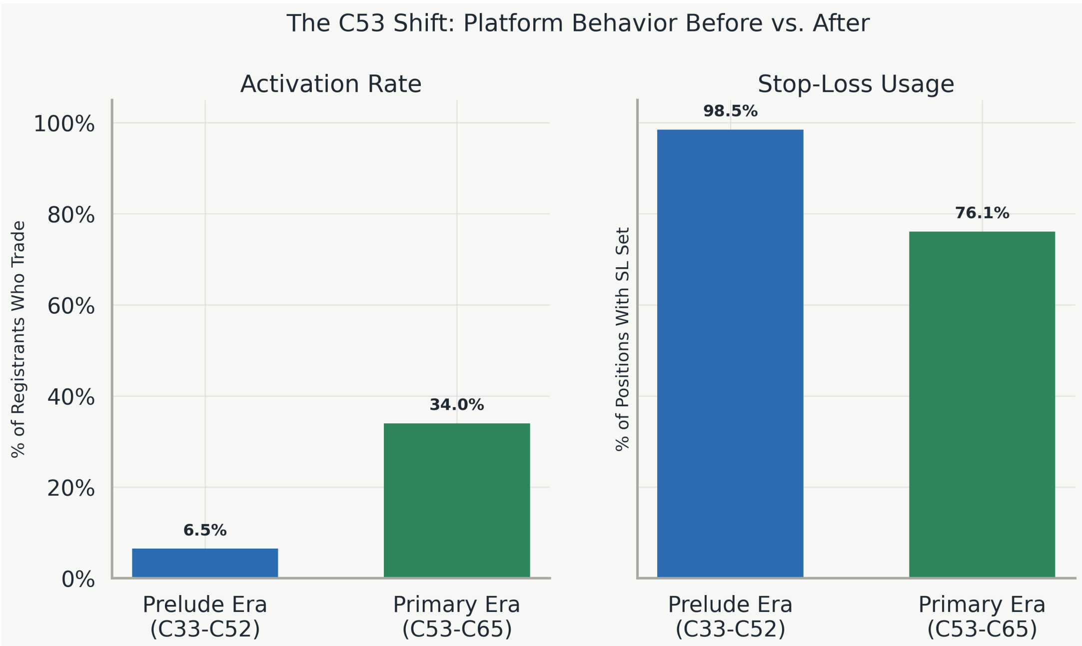
This single fact reframes the modeling objective. Rather than categorizing traders as "good" or "bad," the target state is identifying localized environments where a trader's gross expected loss exceeds the $\$7.00$ per lot hurdle line.

Net economics are overwhelmingly commission-dominated. On the full 34-campaign corpus, mean gross P&L is $-\$1.63$/trade, while median gross P&L is $+\$1.84$/trade — the typical trade is not a loser; a negative-skewed tail of large losses pulls the mean down. Either way, gross trading result is small relative to the $\$41.12$ average commission drag, which accounts for nearly all of the $\$42.75$ average net loss. (This decomposition is corpus-wide and precedes the era discussion below; the $\$7.00$/lot hurdle itself is stable across both eras, so it is presented as a structural fact independent of the behavioral analysis that follows.)



2. The C53 Distributional Shift

The data displays a material structural break at campaign `C53`, justifying the separation of the "Prelude Era" (`C33–C52`) from the "Primary Era" (`C53–C65`) to prevent model pollution.



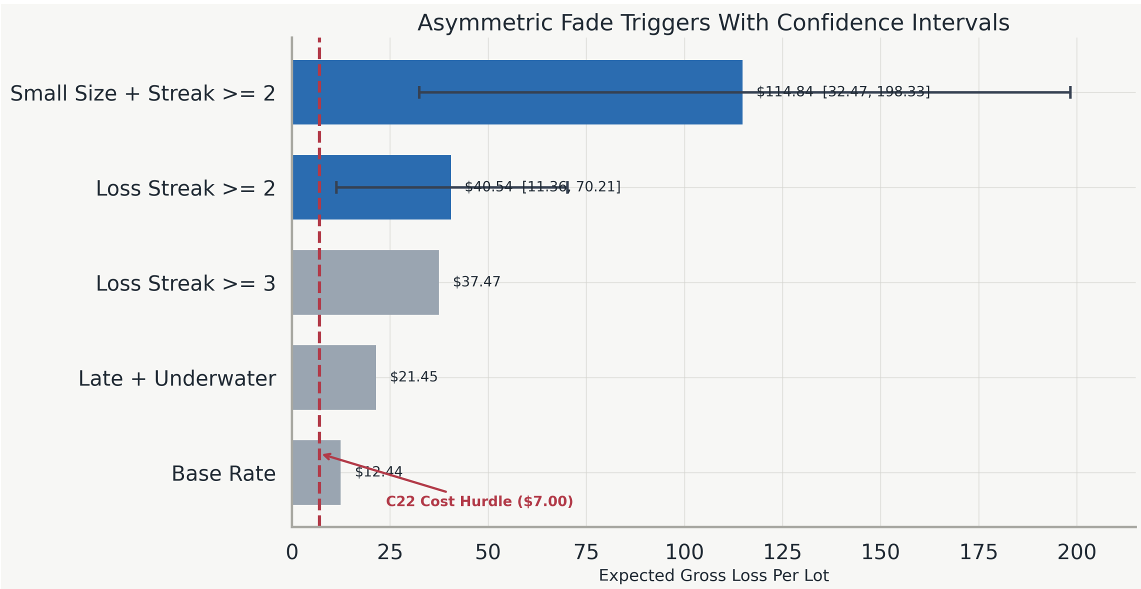
The simultaneous jumps in activation rate and the steep decline in stop-loss discipline indicate a shift in either product mechanics or marketing source. Consequently, all forward-looking behavioral models will be trained and evaluated strictly on the `C53–C65` primary-era window.

2.1 Identity Concentration

Identity concentration in the primary era is high: 68.2% of active traders appear in more than one campaign, and account activity clusters on a small number of shared IPs (up to 141 active accounts on a single anonymized IP cluster). A synchrony test — comparing same-5-minute trade-open overlap within clusters against random account pairs in the same campaign — found no general coordination signature (1.4% vs. 1.0% baseline overlap), consistent with shared infrastructure (e.g. mobile networks, offices) rather than systematic multi-accounting, though 7 of 131 cluster-campaign pairs showed materially elevated synchrony and warrant manual review. Two consequences follow: (1) all statistical tests in Section 3 are clustered by both `traderKey` and `ipClusterId` to bound confidence intervals conservatively even under undetected multi-accounting; (2) C22's stated shared-IP flagging rule may be over-triggering on legitimate shared-infrastructure users while under-triggering on the small number of genuinely synchronized clusters.

3. Behavioral Triggers: Testing H2

Within the primary era, standard linear correlations (such as rank correlation between size and loss) fail to capture localized behavioral blow-ups. However, pre-registered behavioral triggers demonstrate positive gross edge across the board, with two triggers reaching statistical confirmation under conservative clustered resampling. The remaining triggers below are directionally consistent but not independently significant at current sample sizes; they motivate the composite Stage 2–3 model rather than standing alone.



Solid bars with confidence intervals: statistically confirmed under clustered bootstrapping. Lighter bars: directionally positive, not independently significant at current sample size.

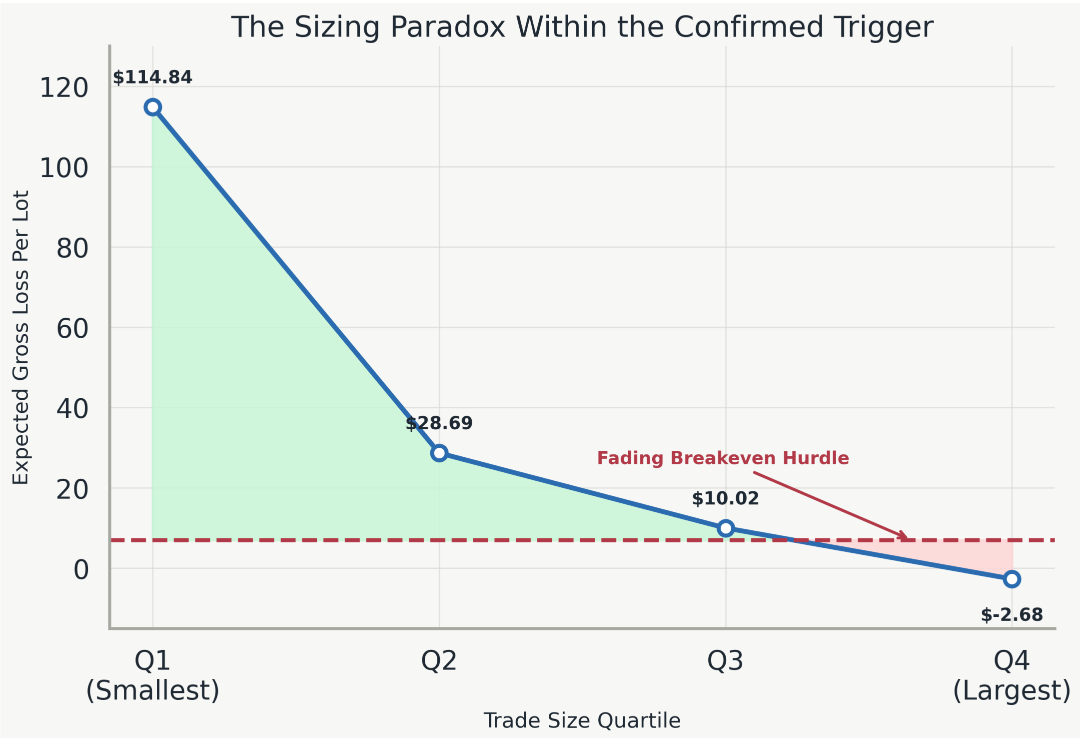
Confirmed trigger 1 — `loss_streak >= 2`. Raw sample mean $\$40.54$/lot (bootstrap-resampled mean $\$40.37$/lot). Clustering by `traderKey` yields a 95% CI of $[11.36, 70.21]$; clustering by `ipClusterId` (to control for unobserved multi-accounting) yields $[10.79, 68.00]$. Both lower bounds clear the $\$7.00$ hurdle.

Confirmed trigger 2 — `loss_streak >= 2 AND small_size_flag` (headline result). Restricting the streak trigger to the smallest size quartile produces the strongest edge in the dataset: $\$114.84$/lot raw mean, with a 95% CI of $[32.47, 198.33]$ (`traderKey`) and $[31.53, 193.50]$

(`ipClusterId`) — more than $16\times$ the cost hurdle, robust to the most conservative identity-clustering assumption tested. This is based on $n = 251$ trades across 150 accounts and 13 campaigns; the effect direction is trusted given its magnitude, but the precise point estimate should be treated as provisional pending Stage 2–3 confirmation on held-out campaigns.

4. The Sizing Paradox

While the gross edge is immensely positive in equal-weighted units, the economics compress materially when weighted by trade amount (equal-weighted mean $\$12.44$/lot vs. size-weighted mean $\$4.00$/lot, primary era). Across the full primary-era population, rank correlation between trade size and per-lot loss is negligible (Spearman $\rho \approx -0.004$) — there is no general size effect. However, *conditional on the confirmed `loss_streak >= 2` trigger*, the relationship becomes sharply and monotonically decreasing:



This quartile decay shows that within the loss-streak trigger specifically, edge is concentrated in the trader's smallest position sizes and disappears (point estimate turns slightly negative) at the largest sizes — consistent with small trades after a streak reflecting forced or tentative risk-taking near account limits, while large trades after a streak may reflect disciplined, system-based re-entries. The practical implication is that **position sizing should enter the fade model as an interaction term with behavioral state, not as a standalone or uniform scaling rule**: neither a flat-size nor a proportional-size fade policy is well-supported as a blanket default, but a size-aware trigger (Section 3, Confirmed trigger 2) is.

5. Data & Integrity Controls

The analyzed corpus covers 34 campaign files containing 46,520 `XAUUSD` fill rows, collapsed to 7,277 unique positions on (`campaignId, positionId`). The primary era (`C53-C65`), excluding single-account test runs, comprises 6,582 positions across 13 campaigns and 496 active accounts.

All raw `email`, `ip_address`, and `telegram_username` fields were aggressively sanitized on ingest. Emails were hashed to `traderKey`, and IP addresses were converted to an anonymous `ipClusterId` via factorization (not hashing, since raw IPv4 space is small enough to be reversible under hashing alone). No raw PII fields are persisted, logged, or utilized in the analytical pipelines.

6. Rule-Design Observations

Two structural observations are relevant to challenge design, offered per C22's invitation to consider how rule and data-generation choices shape outcomes:

- **Effective session length.** Median account active span is 1.99 hours within the nominal 24-hour window (90th percentile: 9.9 hours); most behavioral data is generated well before the window closes. If C22 wants richer late-window behavioral data, mechanisms that extend engagement past the first few hours may increase yield per campaign.
- **Cost-driven vs. skill-driven outcomes.** Since net trader results are overwhelmingly commission-driven rather than skill-driven (Section 1), the challenge's cost structure — not participant ability — is the primary determinant of pass/fail outcomes at the margin. This is a data-generation consideration distinct from the fade-strategy question: a lower-cost or cost-rebated variant might surface more organic (less cost-forced) behavioral signal.